GENERATIVE MODELLING CASE STUDY

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Introduction:

Generative modelling has become one of the important concepts of contemporary data science research, where the emphasis is laid on the discovery of the underlying distributions in the data set and its sample generation based on this distribution. One of the most efficient tools in the context of generative models is Generative Adversarial Networks (GANs).
GANs consist of two neural networks:

- A **Generator**, which produces synthetic data.
- A **Discriminator**, which evaluates whether data is real or generated.

These networks are trained in an adversarial manner, where the generator attempts to fool the discriminator while the discriminator attempts to correctly classify inputs.

The aim of the research is to:

- Understand the conceptual application of GANs using artificial data.
- Using GANs in various fields.
- To Evaluate their effectiveness using visual and numerical methods.

The research will be subdivided into two parts:

- Part 1 – Artificial data modeling
- Part 2 - Application in reality:
    - medical field (BloodMNIST)
    - cybersecurity (CICIDS 2017)
    - creative field (QuickDraw sketches)

Theoretical Background:

GAN Framework:

The Generative Adversarial Network (GAN) comprises two neural networks, which are trained concurrently using adversarial learning techniques: the generator and the discriminator. The objective of the generator is to create data samples similar to the actual data samples, whereas the discriminator tries to differentiate between the two.

The generator takes as input a random noise vector $z \sim p_z(z)$, typically drawn from a Gaussian or uniform distribution, and transforms it into a data sample $G(z)$. The discriminator receives both real data $x \sim p_{data}(x)$and generated data $G(z)$, and outputs a probability representing whether the input is real.

The training process is formulated as a minimax optimisation problem:

$$min\ _G\ max\ _D\ V(D, G = E_\{x \sim p_data(x)\ \}[log\ D(x)\ ] + E_\{z \sim p_z(z)\ \}[log\ (1 - D(G(z)))\ ]$$

In this formulation:

The discriminator tries to maximize the chances of accurately discriminating between real and fake samples. The generator tries to minimize the chance that it can be classified as fake, through this game of cat and mouse, the generator is pushed by the discriminator to get better until it produces realistic data. In theory, at equilibrium, the generator will learn the actual data distribution, and the discriminator will classify any input with a probability of 0.5.

Unfortunately, reaching this equilibrium proves difficult due to problems related to gradients and complex distributions.

Training Challenges:

GAN training is inherently unstable due to:

- non-convex optimisation
- simultaneous training of two networks
- adversarial dynamics

Common issues include:

- Mode collapse: generator produces limited variety
- Vanishing gradients: discriminator becomes too strong
- Training oscillations

Metric Evaluation: Fréchet Inception Distance:

To measure the quality of images produced, Fréchet Inception Distance (FID) metric is applied. This approach does not involve comparing images at the pixel level but calculates the distance between the distribution of real and generated images in the feature space.

For the extraction of high-level image features, a pre-trained Inception neural network is employed. The features include semantics such as shape and texture and do not represent raw pixel values.

The distance between the two distributions is defined as:

$$FID = \left\|\mu_r - \mu_g\right\|^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r \Sigma_g\right)^{1/2}\right)$$

Where:

- $\mu_r$and $\Sigma_r$are the mean and covariance of real image features.
- $\mu_g$and $\Sigma_g$are the mean and covariance of generated image features.

The first quantity gives the difference between means, while the second quantity gives the difference between covariances. The smaller the FID, the better because it signifies that the distribution of the generated samples is like the real sample's distribution. Nevertheless, FID is sample size-dependent, hence the use of fewer samples in the current study.

Synthetic Data Modelling

Sine Wave GAN

To develop an intuition behind the behavior of a GAN, a synthetic data set was first created using a sine wave as data. This data set is low-dimensional and structured in nature, which allows the GAN to map its latent space into its data space easily. The architecture of the generator was such that the input was a random variable z~N(0,1), while the output was a 2D point coordinate. The architecture of the discriminator on the other hand, comprised an MLP tasked with discriminating between actual samples of a sine wave and generated ones. The two were trained through iterations based on the adversarial loss function stated above.

Results and Observations:

- The produced data followed the sinusoidal function closely.
- The behavior of the loss functions depicted consistent convergence.
- There was no sign of domination of the discriminator over the generator.

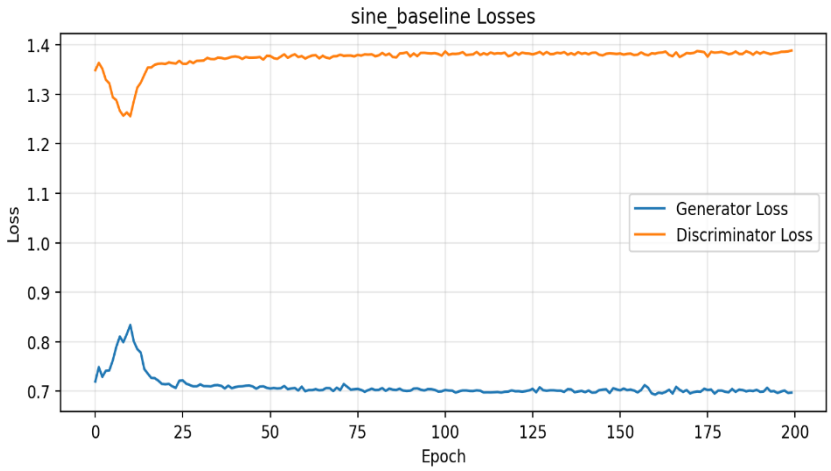

Figure 1: Generator and Discriminator Loss for Sine Wave GAN

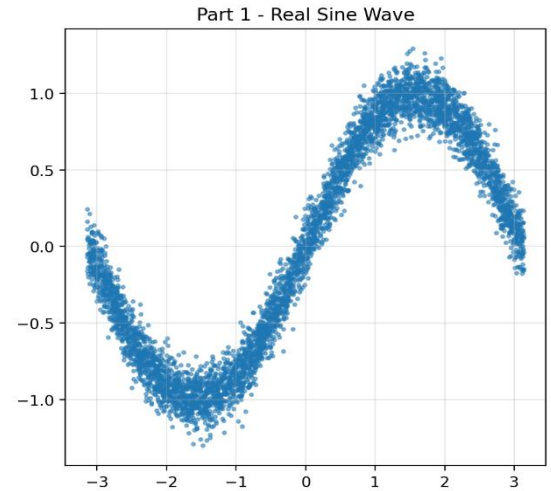

Figure 2: Generated Sine Wave using GAN

It can be observed that GANs are extremely powerful tools when working with simple distributions. The simplicity and low dimensionality of the distribution made it easy for the generator to capture the mapping function.

Mixture of Gaussians

To create complexity in the data set, a combination of Gaussian distributions was employed. This data set has multiple modes, which makes it harder for the GAN to learn the entire distribution.

There were two architectures that were employed:

- A baseline GAN architecture
- A modified GAN architecture with extra hidden layers

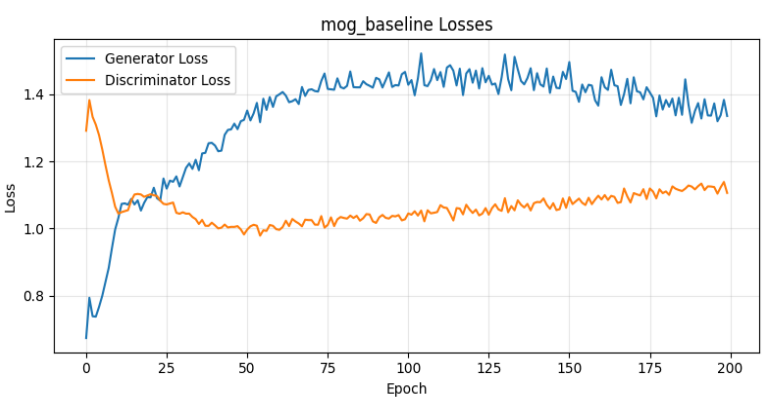

Figure 3: Real Mixture of Gaussians

Results and Observations:

- Baseline network had difficulty capturing all the clusters in Gaussian.
- Samples generated by the network clustered themselves around a few modes.
- Architecture improvement helped with better capturing of clusters.
- But some degree of mode collapsing could not be avoided.

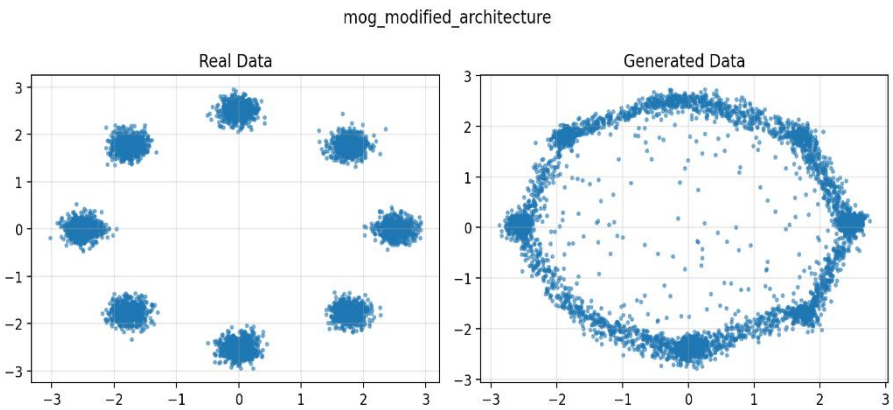

Figure4 MOG Real and Fake

Analysis:
Mode collapse is one of the key drawbacks of GAN models; this can be seen from the results of this experiment because the generator cannot model all modes of the input distribution even when the complexity of the model increases. In conclusion, it should be noted that GANs work very well in controlled conditions but fail when facing a more complicated input distribution.

BloodMNIST (Medical Imaging)

Dataset Exploration

The BloodMNIST dataset consists of labelled microscopic images of blood cells. The images were preprocessed by normalising pixel values and ensuring consistent input dimensions for the DCGAN model. A class distribution analysis was performed to understand the balance of the dataset.

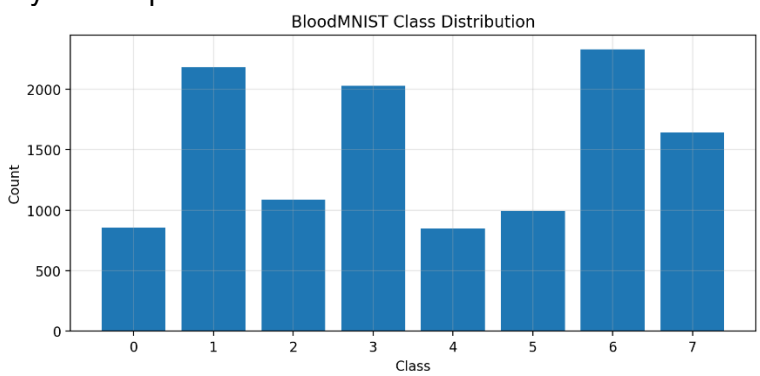

Figure 5: BloodMNIST class distribution

The dataset shows a reasonably balanced distribution, making it suitable for training a generative model**.**

Model Architecture

The model used in this work for this task is DCGAN which stands for Deep Convolutional GAN. The convolutional layers were applied both in the generator and the discriminator network for identifying features like edges and textures. The generator gradually upscales the input latent vectors to images, whereas the discriminator downscales images to probabilities.

Training Results

During training, the generator and discriminator losses were monitored.

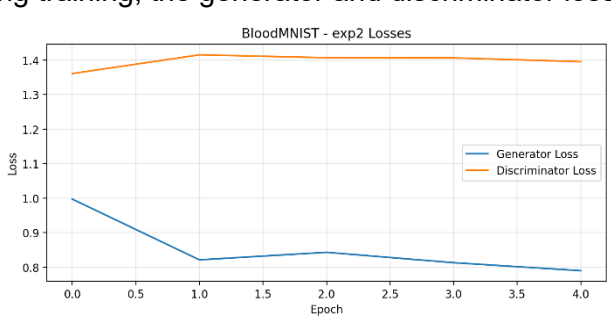

Figure 6: BloodMNIST Exp 2 Loss

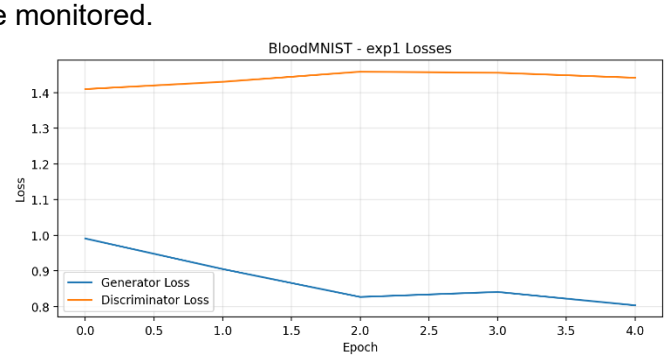

Figure 7: BloodMNIST Exp 1 Loss

Observations:

- Generator loss decreased steadily, indicating improved sample quality.
- Discriminator loss remained relatively high, suggesting ongoing competition.
- The adversarial balance was maintained throughout training.

Generated Images

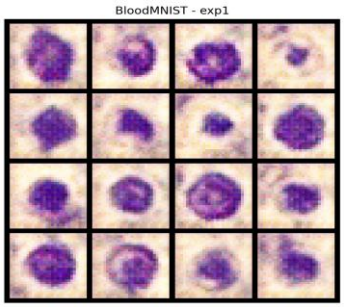

Figure 8: Generated BloodMNIST Sample 1

The generated images exhibit:

- realistic colour distributions
- identifiable cell-like structures
- blurred boundaries and limited fine detail

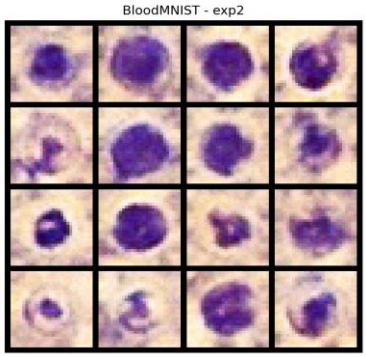

Figure 9: Generated BloodMNIST Sample 2

Analysis

While the model was able to learn coarse characteristics like shapes and colours, it had difficulty learning fine-level characteristics like textures and edges.
This shows that:

- The generator was able to capture low frequencies,
- It was unable to learn high frequencies.

This is a typical problem associated with training GANs, particularly when there are few epochs.

Hyperparameter Comparison

Two experiments were conducted with different batch sizes.

Experiment	Batch Size	Epochs	Generator Loss	Discriminator Loss
Exp1	128	5	~0.80	~1.44
Exp2	64	5	~0.84	~1.45

The results indicate that:

- smaller batch sizes slightly improved diversity
- training stability remained similar across configurations

Evaluation (FID)

Calculation of the Fréchet Inception Distance was done through a selected subset owing to computational limitations. Despite such limitations, FID is quite an effective measure for supporting visual results.

Cybersecurity (CICIDS 2017)

Dataset Processing

CICIDS 2017 dataset was chosen for modelling network traffic behaviour. The Wednesday dataset was considered, specifically the BENIGN attacks and DoS.

The following operations were performed on the data:

- Cleaning
- Filtering
- Normalisation

Each data point consists of a high-dimensional vector of features.

Model

A GAN with fully connected layers was used for this tabular data. Unlike image-based GANs, this model does not leverage spatial relationships, making the task significantly more challenging.

Observations:

- Generator loss decreased slowly
- Discriminator remained dominant
- Training convergence was weaker compared to image tasks

Distribution Comparison

PCA and t-SNE were used to visualise real and generated distributions.

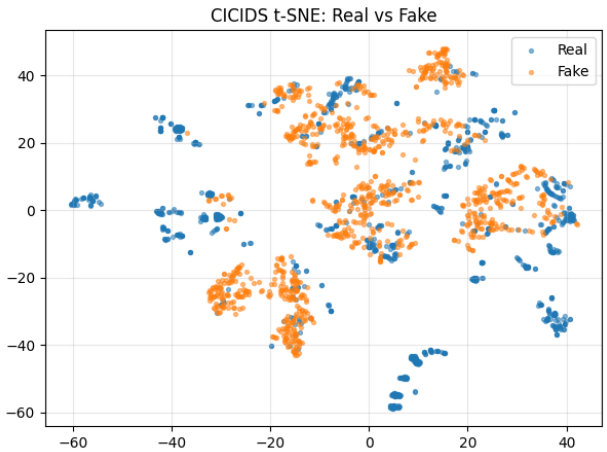

Figure 10: t-SNE Real vs Fake

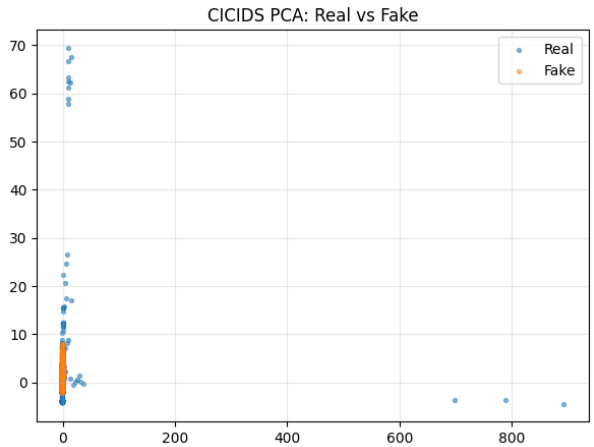

Figure 11: PCA Real vs Fake

Feature-wise comparisons were also made.

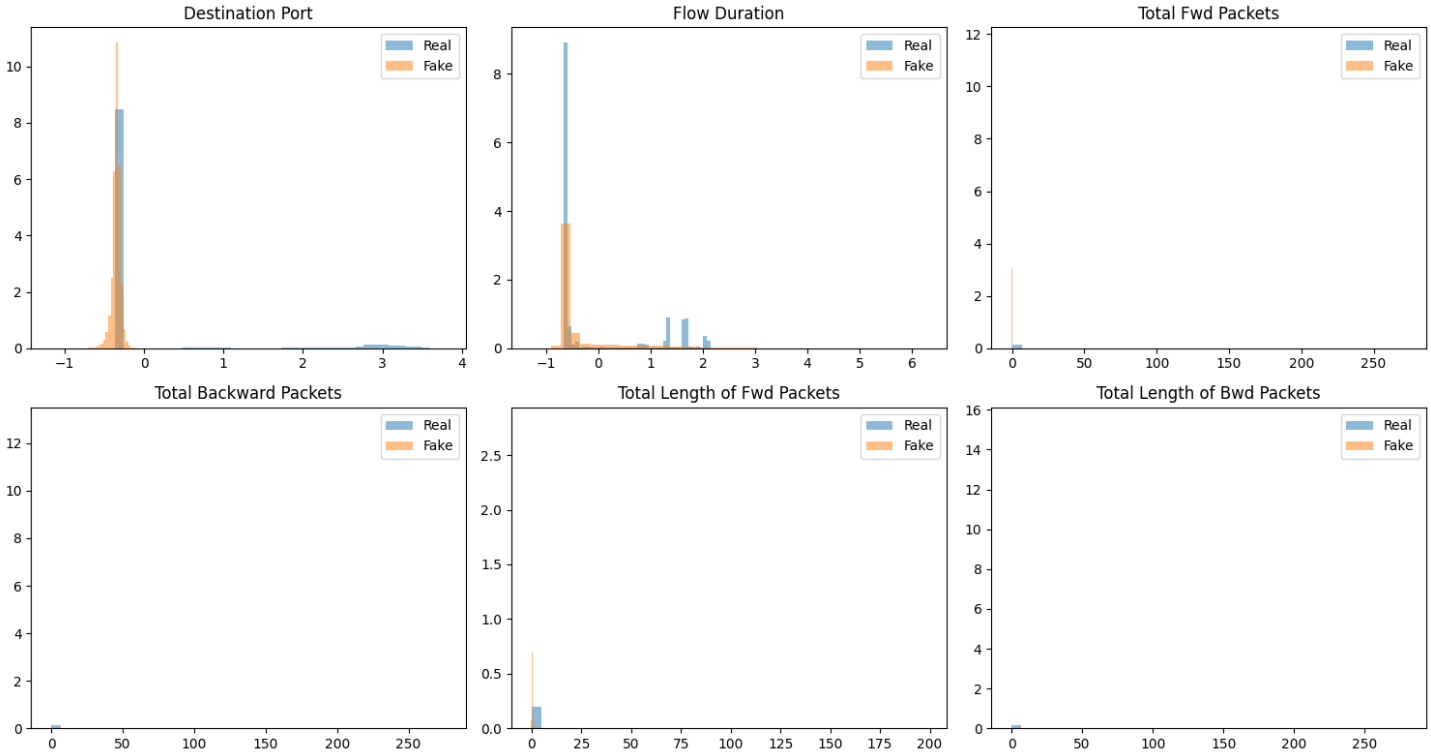

Figure 12: Histogram Comparison

Analysis

The data produced showed some overlap with actual data but was not diverse.
Important observations:

- Samples produced were centered on mean values.
- Dispersion was much smaller than actual data.
- Feature distribution was not completely modeled.

Interpretation

This proves that GAN models cannot handle:

- high dimensional feature space
- complex correlation
- no spatial information

The GAN model was found to suffer from mode collapse; it could only produce few variations.

QuickDraw (Creative AI)

Dataset

The QuickDraw Dataset comprises basic drawings. The ‘pizza’ class is chosen, and images were converted into 28x28 gray-scale images.

Model

A DCGAN architecture was used, like the BloodMNIST experiment. Due to computational constraints, a reduced subset of the dataset was used.

Results

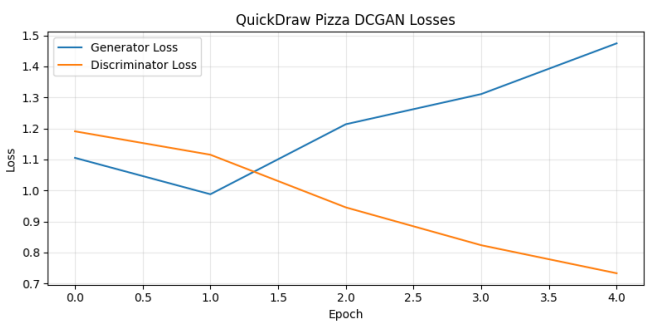

Figure 13: QuickDraw Pizza DCGAN Loss

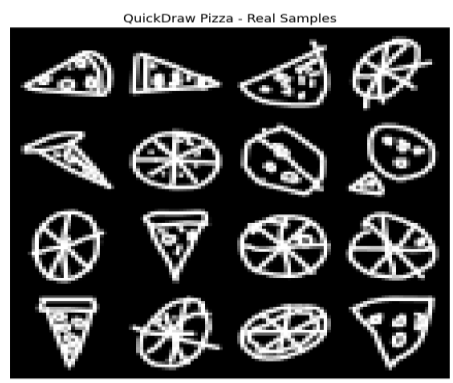

Figure 14: QuickDraw Pizza - Real Samples

Analysis

The generated sketches illustrate:

- Distinct circles
- Similarity of toppings
- Irregularity in the pattern

It shows that the model can capture the general structure but not the finer aspects.

Evaluation

FID was calculated using only a subset of the samples.

The higher FID score suggests:

- Imperfect training convergence
- Insufficient training data
- Sketch irregularity

Discussion

The experiments provide important lessons on GAN performance within various contexts.

- GANs work well for low-dimensional and structured data.
- Image-related problems can be solved by spatial learning.
- Tabular problems continue to pose challenges due to their complexity.

The findings also demonstrate the significance of:

- Model structure
- Hyperparameters
- Computational capabilities

Conclusion

This paper examined the usage of GANs in synthetic data, medical image synthesis, cybersecurity, and sketch drawing applications. It was found that GANs can perform exceptionally well for modelling structured and low-dimensional datasets, such as when generating sine waves and Gaussian mixture models. In both cases, the GAN generator managed to learn from the distribution and produced output like real datasets.

The visual models developed for the BloodMNIST and QuickDraw could produce visually understandable results, identifying crucial aspects such as shapes and colors. The images produced by these models did not have finer details, suggesting that the models had successfully learnt the low frequency aspects but failed to understand the high-frequency aspects. The reason behind this can be attributed to insufficient time and processing power.

On the other hand, the CICIDS experiment proved that GANs face considerable difficulties in dealing with high dimensional tabular data. Limited variability was observed as well as inability to reflect the dependencies between features, which led to mode collapse and lower variance in the data. Generally, the results indicate that GANs can be very efficient with images and structured data, but their success largely depends on the type of data and its properties. It would be interesting to investigate the potential of more complex neural networks like WGANs and cGANs.

Furthermore, the experiments show the necessity of designing and evaluating carefully a good model when using generative modelling. Visual comparison, along with metrics like FID, played an important role in measuring the effectiveness of the model since numbers cannot always reflect quality. Furthermore, the experiments stress the need to have balanced training of the generator and discriminator of the GAN architecture; otherwise, poor results may be obtained due to model instability. In practice, the paper shows that even though GANs are capable of doing amazing things, its success hinges on proper preparation, tuning, and computing power.